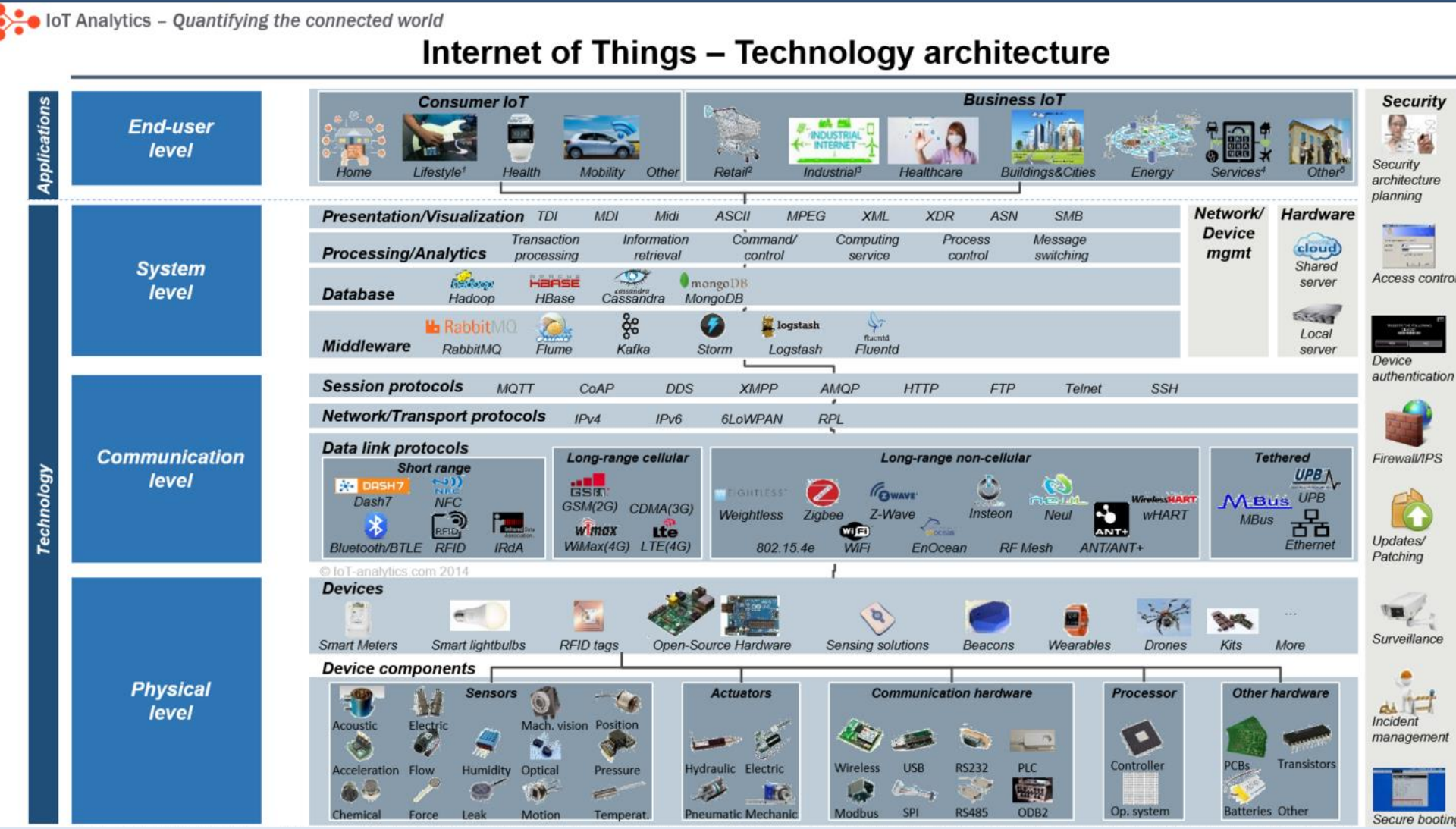
SAKARYA ÜNİVERSİTESİ
Bilgisayar ve Bilişim Bilimleri Fakültesi
Bilgisayar Mühendisliği Bölümü

BSM 451
INTERNET OF THINGS AND APPLICATIONS

IoT Wireless Communication Technologies
Bluetooth Low Energy (BLE)

Assoc. Prof. Cüneyt BAYILMIŞ
Researcher Dr. Ünal ÇAVUŞOĞLU

IoT Technology Architecture



IoT Wireless Communication Technologies

- Short Range
  - Radio-Frequency Identification, RFID,
  - Near Field Communications, NFC,
  - **Bluetooth Low Energy (BLE),**
  - Infrared (IRdA)
- Non-Cellular Long-Range
  - Wireless Sensor Networks (IEEE 802.15.4 - ZigBee),
  - Z-Wave
  - ANT/ANT+
- Cellular Long-Range
  - GSM / GPRS
  - 3G / 4G (LTE) / 4.5G (LTE Advanced)
  - WiMAX

Bluetooth

- Bluetooth is a short distance, fast, secure, reliable, low-cost wireless network model for personal, office, industrial areas.
- Bluetooth is an open standard and free to use technology.
- Basic Bluetooth standard supports short distance (~10m) upto 1 Mbps data transfer in licence-free 2.4 GHz ISM (Industrial, Scientific, Medical).
- Bluetooth 1.0 was introduced in 2001, Bluetooth 1.1 in 2002, Bluetooth 1.2 in 2003.
- Bluetooth 2.0 and 2.0 EDR versions support data transmission rates of 2 Mbps and 3 Mbps, respectively and they both are compatible with older versions.
- Bluetooth 3.0 was introduced in 2009 and, differently from older versions, it provides low power consumption and 802.11 is added as fast data transfer.
- Bluetooth 4.0 version's spotlight feature is high security with 128-bit AES encryption.



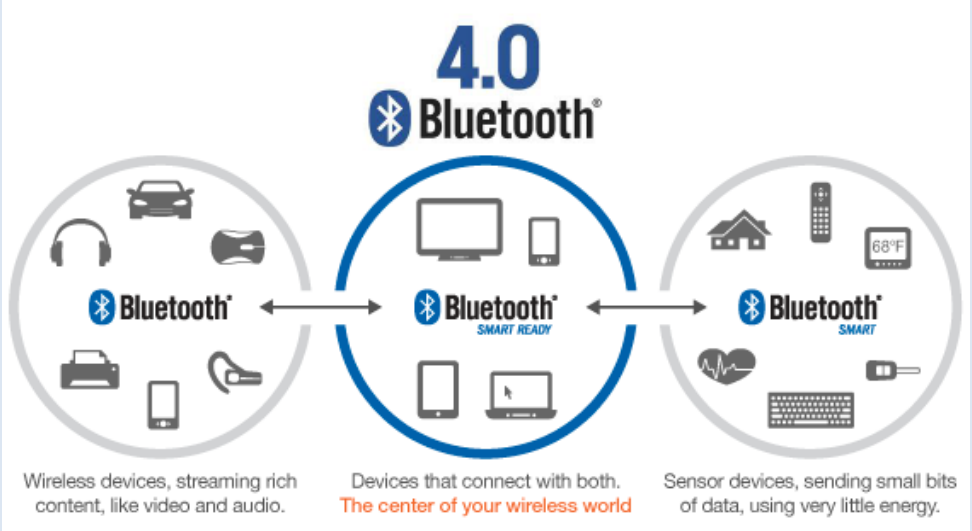
Fact : Bluetooth is considered as the fundemantal of IEEE 802.15 standard family.

Bluetooth Network Architecture

- It supports peer to peer, ad-hoc topologies considering close proximity networks.
- Two devices can communicate automatically once they are in their coverage areas.
- Master / Slave architecture. Master's task is to manage synchronization of , Frequency Hopping Spread Spectrum, FHSS communication between Bluetooth devices.
- The smallest network unit formed by communicationg Bluetooth devices is called Piconet. When two or more Piconets partially over each other, it is called Scatternet.
- A Piconet can be formed by upto 8 active devices, one has to be Master and other have to be Slaves. Master can also connect 255 Slaves in stand-by mode.

Bluetooth Low Energy (BLE)

- aka Bluteooth smart / Bluetooth 4.0.
- Same coverage area, low power consumption.
- Specifications
  - Frequency Hopping Spread Spectrum, FHSS,
  - 2.4 GHz frequency band (Industrial Scientific Medical, ISM),
  - Time Division Multiple Access, TDMA,
  - 1 Mbps transmission rate,
  - Scalable to 5917 slaves,
  - 0.01 – 0.5 w power consumption,



Bluetooth Technology

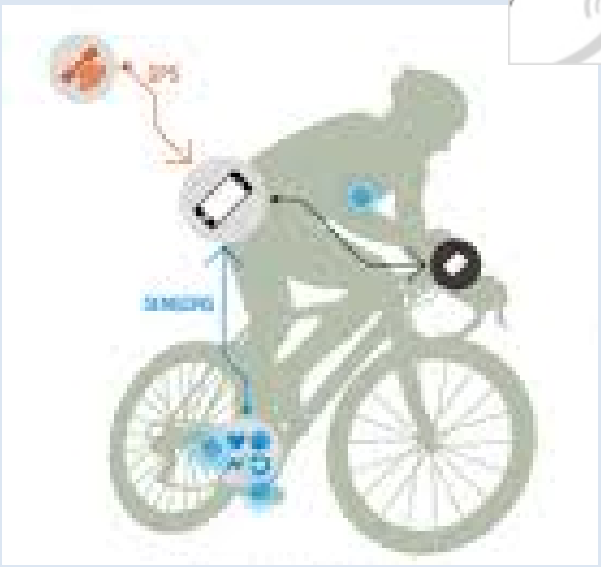
Technical Specification	Classic Bluetooth technology	Bluetooth low energy technology
Radio Frequency	2.4 Ghz	2.4 Ghz
Distance/Range	30 meters	50 meters
Over the air data rate	1–3 Mbit/s	1 Mbit/s
Application throughput	0.7–2.1 Mbit/s	0.2 Mbit/s
Active slaves	7-16,777,184	Unlimited
Security	64/128-bit and application layer user defined	128-bit AES and application layer user defined
Robustness	Adaptive fast frequency hopping, FEC, fast ACK	
Adaptive fast frequency hopping		
Latency (from a non-connected state)	Typically 100 ms	6 ms
Total time to send data	100 ms	<6 ms
Government Regulation	Worldwide	Worldwide
Certification Body	Bluetooth SIG	Bluetooth SIG
Voice capable	Yes	Yes, with some limitation
Network topology	Scatternet	
Star-bus		
Power consumption	1 as the reference	0.01 to 0.5 (depending on use case)
Peak current consumption	<30 mA	<15 mA
Service discovery	Yes	Yes
Profile concept	Yes	Yes
Primary use cases	Mobile phones, gaming, headsets, stereo audio streaming, automotive, PCs etc.	Mobile phones, gaming, PCs, watches, sports and fitness, healthcare, security & proximity, automotive, home electronics, automation, Industrial, etc.

http://rtcmagazine.com/articles/view/103741

Bluetooth Low Energy (BLE)

- BLE provides seven profiles.
  - Health care profiles
  - Sports and fitness profiles,
  - Internet Connectivity,
  - Generic Sensor,
  - HID Connectivity,
  - Proximity sensing,
  - Alert and time profiles,





Bluetooth Low Energy (BLE)

- BLE vs. other short-range wireless technologies

Wireless tech.	Transmission rate	Frequency	Range	Avg. Network lifecycle
BLE IEEE802.15.1 (v4)	1 Mbps	2.4GHz	10-50 m	1-2 years
Bluetooth IEEE802.15.1	1-24 Mbps	2.4GHz	10-100m	days – months
UWB IEEE802.15.3a	110–480 Mbps	3.1 -10.6 GHz	10 m	1 – 2 years
ZigBee IEEE802.15.4	20–250 Kbps	2.4GHz	100 m	6 months – 1 year
Z-Wave	40 Kbps	868/908MHz 2.4 GHz	100 m	1 year

Kaynak: C. Bayılmış, M. Özdemir, "Bluetooth Düşük Enerji Teknolojisine Sahip İşaretçi ve Akıllı Telefon Temelli Öğrenci Yoklama Sistemi", Gazi Üniversitesi Bilişim Teknolojileri Dergisi, 9 (3), 249–254, 2016.

Beacon: BLE Tech. Based Pointer Devices

- Beacons are low power-low cost electronic transmitters decoreated with BLE technology.
- Can communicate with Android ve iOS devices.
- Beacon devices broadcasts signals called BLE Advertising.
- Mobil devices react (run an application) when they receive signals from Beacon devcies.
- Therefore, applications which requires micro-geolocation, actitivities realted to location can be realized.



Beacons

iBeacon



Blesh

Beacon Devices



Estimote

Eddystone

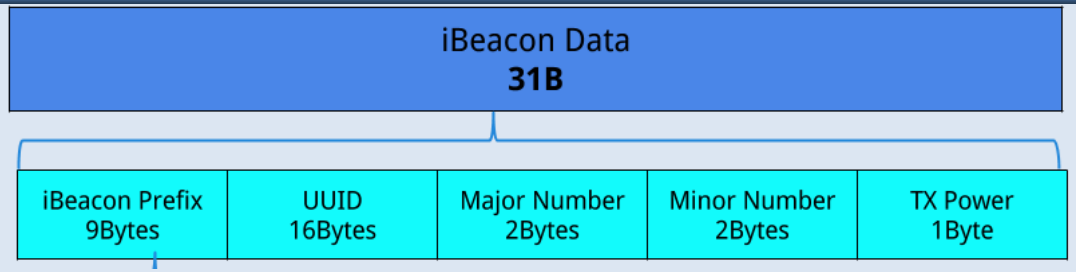
Google



Phoneist

- Bluegiga BLE 113 chip used in some Beacon devices consumes max.18,2 mA while transmitting, max 14.3 mA while receiving, and 0,4 μA in stanby mode.
- Some companies that develops beacon devices
  - Apple: iBeacon for developers, https://developer.apple.com/ibeacon/
  - Estimote beacon, https://estimote.com
  - Google beacon eddystone, https://developers.google.com/beacons/
  - Blesh beacon, https://www.blesh.com
  - Boni beacon, https://www.boni.me
  - Phoneist beacon, https://www.phoneist.com

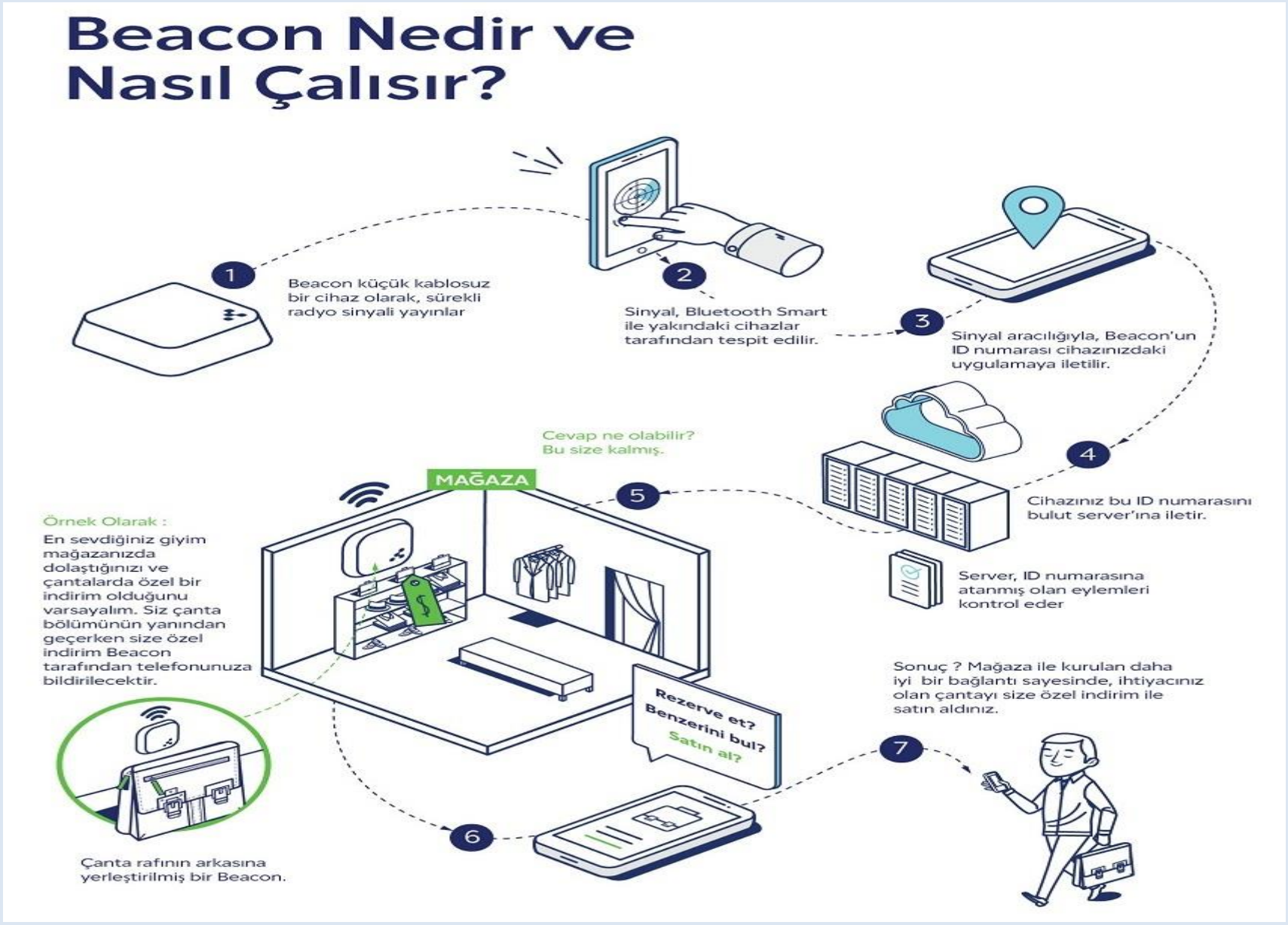
iBeacon Packet Format



- Bluetooth signals broadcasted from beacon devices carry four types of data.
- Universal Unique Identifier, UUID
  - Used to define whole beacon network (manufacturer ID)
  - 16 bytes length.
  - For instance, when a mobile application senses an UUID that belongs to it, it starts to listen the rest of the message.
- Major
  - Upper network address. 2 byte length.
- Minor
  - Lower network address. 2 bytes length.
  - Shows related device in the same upper network.
- TX Power
  - RSSI (Received Signal Strength Indication) data to determine distance of beacon device. (RSSI data in 1m)

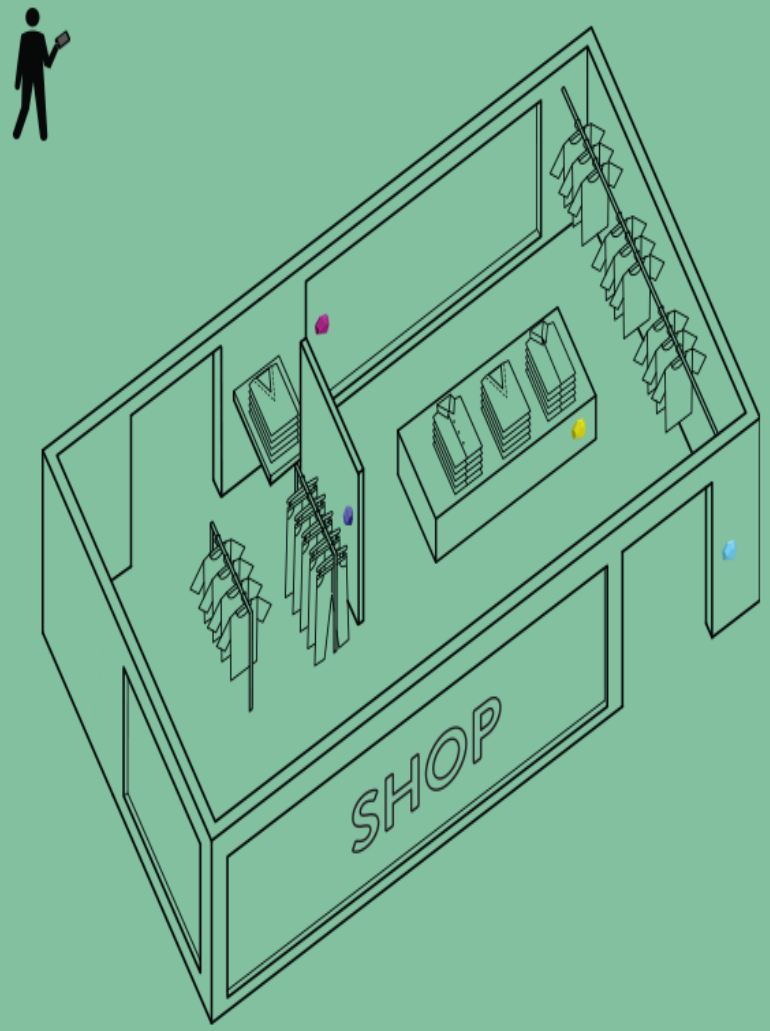
Example: Beacons in faculty building: Faculty building number is UUID, first floor is major 1 and classroom 1003 is minor 03.

What is beacon? How does it work?



Beacon Application

Using Beacons – an example retail implementation



John is headed to his favorite retail store. He has the store's mobile app on his smartphone and although it's in his pocket, it's picking-up signals from beacons located in the store.

Beacon Application

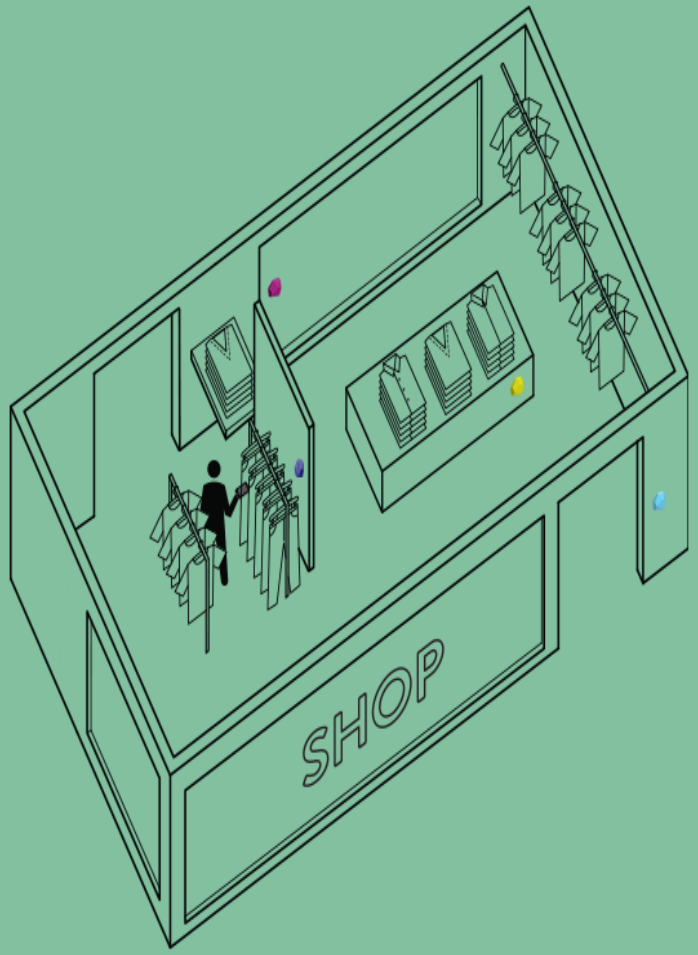
Using Beacons – an example retail implementation



As John approaches the entrance, the violet beacon positioned in the window, sends a personalized message to his phone engaging him to read the welcome message.

Beacon Application

Using Beacons – an example retail implementation



As John approaches the entrance, the violet beacon positioned in the window, sends a personalized message to his phone engaging him to read the welcome message.

Beacon Application

Using Beacons – an example retail implementation



John sees some great jeans and wants to learn more about them. Thanks to another beacon located near the product, he can immediately receive more information and a special offer.

Beacon Application

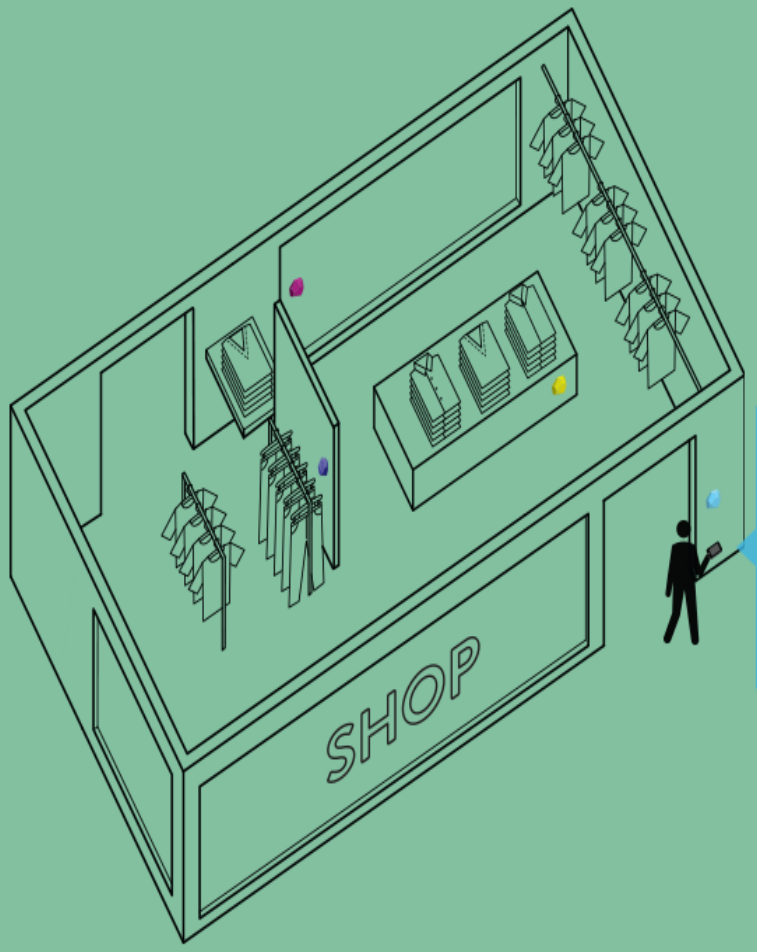
Using Beacons – an example retail implementation

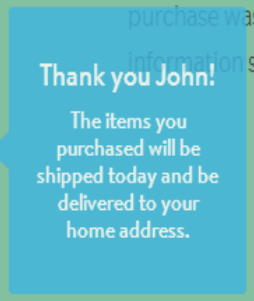


As he approaches the t-shirt section of the store the yellow beacon sends a new message to John's phone with a recommendation for a new purchase. John decides to buy both of the items and uses the store's mobile app to finalize the purchase.

Beacon Application

Using Beacons – an example retail implementation



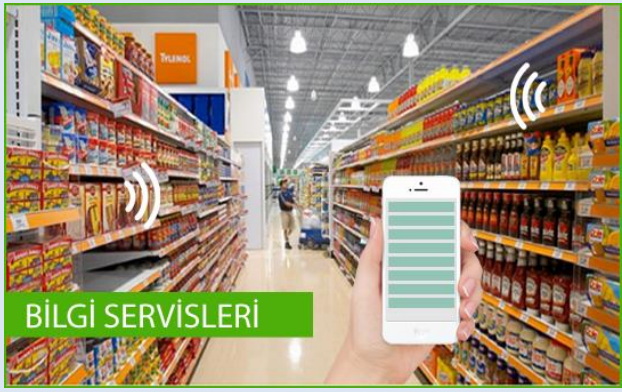


As John exits the store the light blue beacon triggers a thank you message with shipping information which appears on his phone. His purchase was completed with the credit card information stored in the store's mobile app.

Beacon Applications

- Retail stores, supermarkets
- Museums, historical places, attractions,
- Attendance,
- In-door navigation
- Banking







Beacon Eddystone

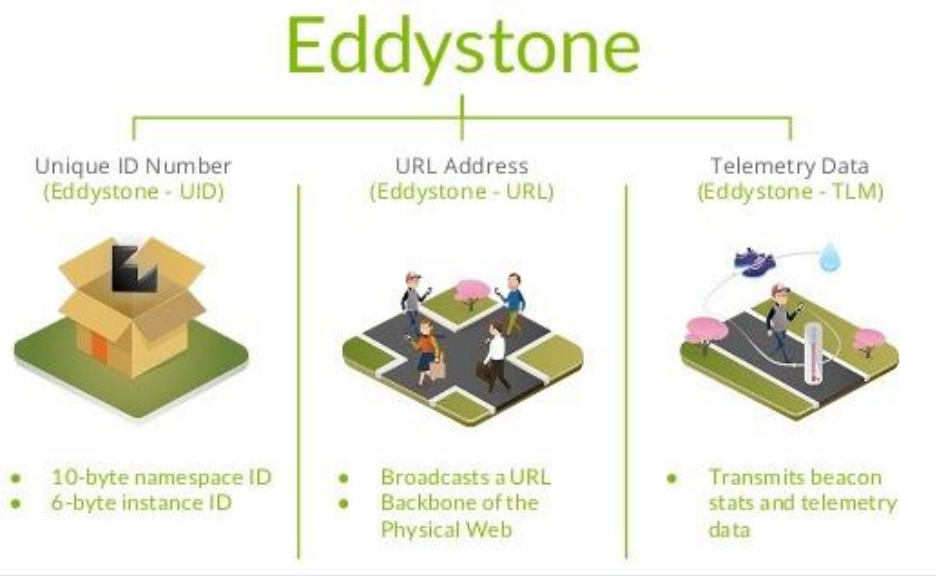
- Eddystone developed by Google is an open source and multi-platform (Android ve iOS) BLE Beacon format.
  - https://github.com/google/eddystone
- Introduced in July 2015.
- Google Eddystone is open source platform for companies manufacturing beacon devices.
- Google Eddystone can run in many becaon devices manufactured by different companies (Estimote, Kontak.io, Blesh, Radius Networks [Altbeacon] vd.).
- ibeacon and Eddystone are used for broadcasting identification packets similar to BLE devices.

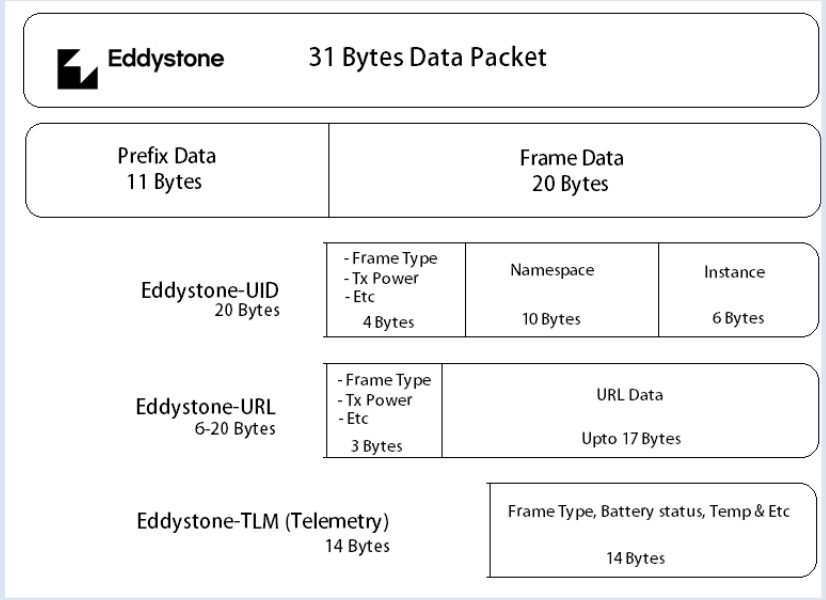


Note: Eddystone's main difference from other BLE Beacons is URL data sent in BLE packets.

Packet Format

- Eddystone has three different packet types (UID,URL,TLM) and supported in both iOS and Android by Google.
  - UID is similar to beacon UUID. Two parts: 10 bytes beacon identification data (Namespace) and 6 bytes data for group or manufacturer differentiation.
  - Applications are directly sent to URL by URL packet (17 bytes). (Compressed URL)
  - TLM (telemetry) includes data gathered by battery, temperature, humidity, etc. sensors.

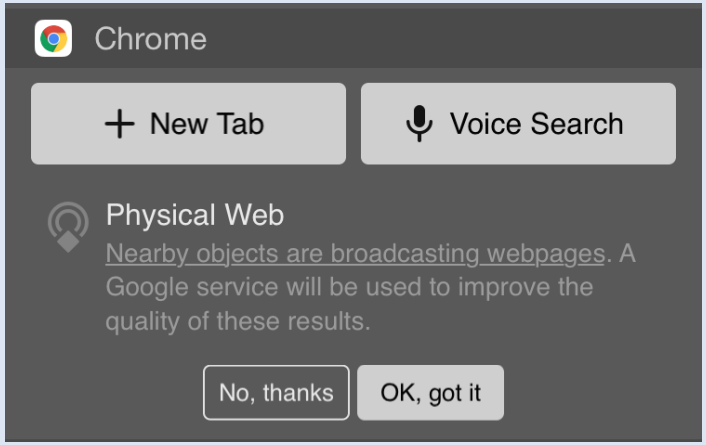


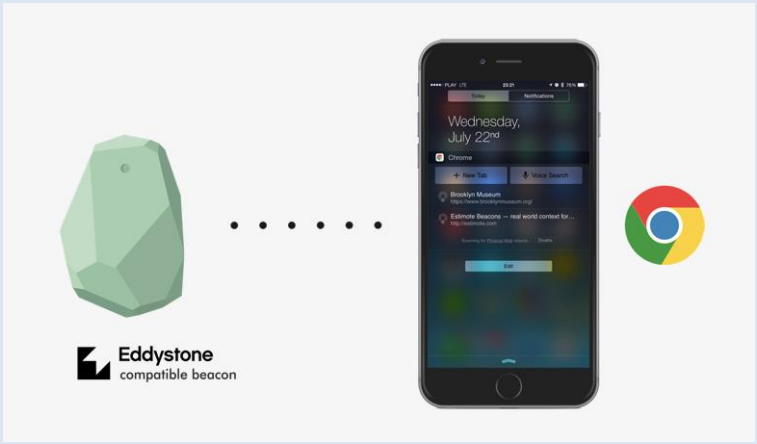


Physical Web

The Physical Web

- Controlling and communicating with BLE tech devices (beacons) via web browsers (Chrome) without downloading a mobile application.
- Google takes adavantage of Eddystone-URL packet.





iBeacon vs. Eddystone





Features	ibeacon	Eddystone
Platform Support	Only iOS	Android and iOS
Packet formats	Single type (UUID)	Three types: UID, URL, TLM
Mobile application requirement for end user	Yes	No (thanks to Eddystone-URL packet format)

Receiving Beacon Messages by an Android App

- Receiving Beacon messages by an Android application:

```
…BeaconManager.getInstance(context).registerRangingListener(new …BeaconListener() {

   @Override
   public void onNearestRangedBeacon(…Region …Region,
                                     …Beacon …Beacon)
```

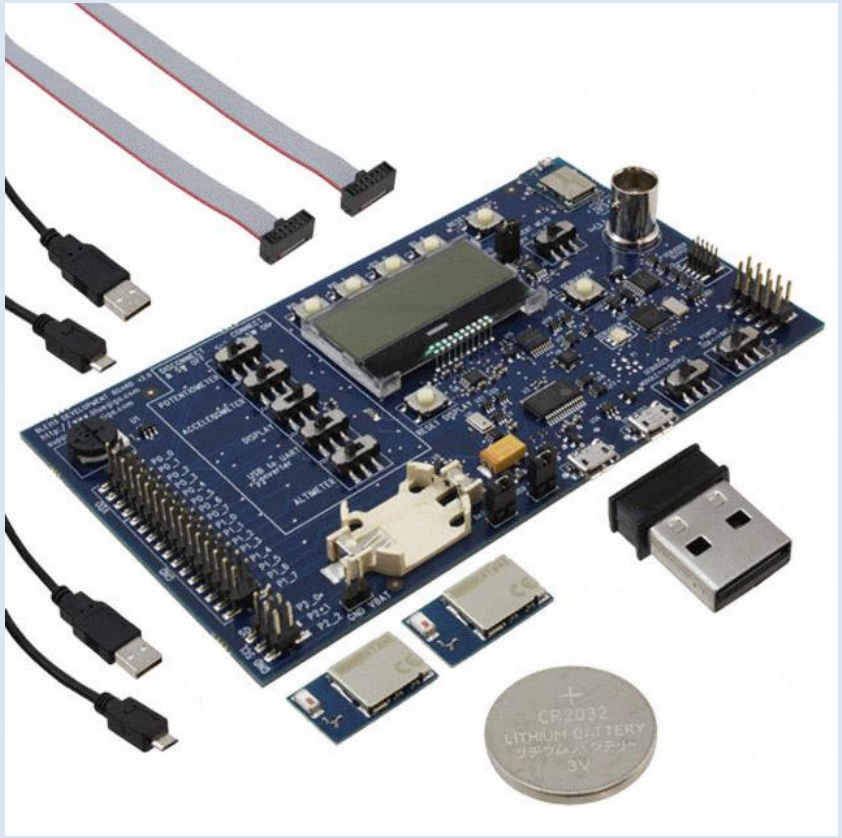
!!! … refers to Beacon manufacturer company name.

- …With an object created in Beacon class, Beacon information is obtained. This should be done in onNearestRangedBeacon method.
- …Beacon.getMinor() : get minor address of beacon of which signals received.
- …Beacon.getRssi() : determine the signal strength of beacon.

How to program Beacon devices?

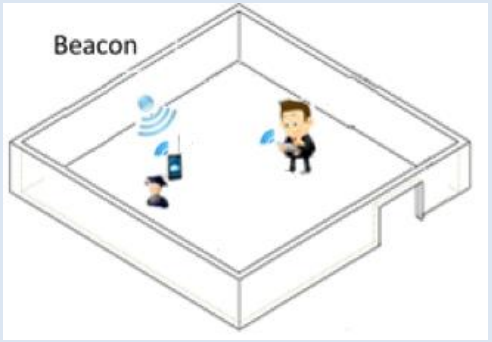
- Major, minor addresses, Tx signal strength can be adjusted by Developer kits or programmer.

- To program Bluegiga BLE113 chips: DKBLE113 developer kit.



Application examples: Beacon Micro Positioning Program

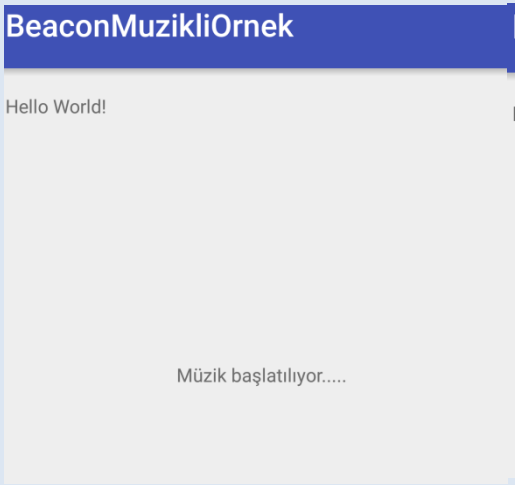
- When you enter the coverage area of the Beacon device and the application that sends notification or plays music.

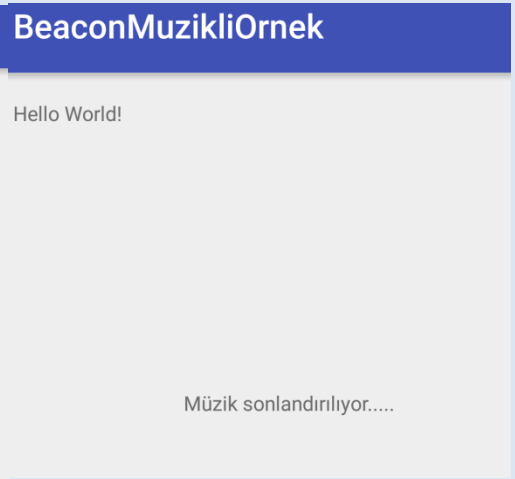


Sender Application

Öğrenci Adı : Ahmet
Soyadı: YILMAZ
Tarih : 06/02/2017
Giriş Saati : 16:50:25
Çıkış Saati : 16:51:22

Music Playing Application





Uygulamayı hazırlayan Fatma KOÇ

Lucky Beacon

- The user entering the business in a certain order (beacon coverage area) is considered as a lucky user and no gifts or discounts are given to the related user.

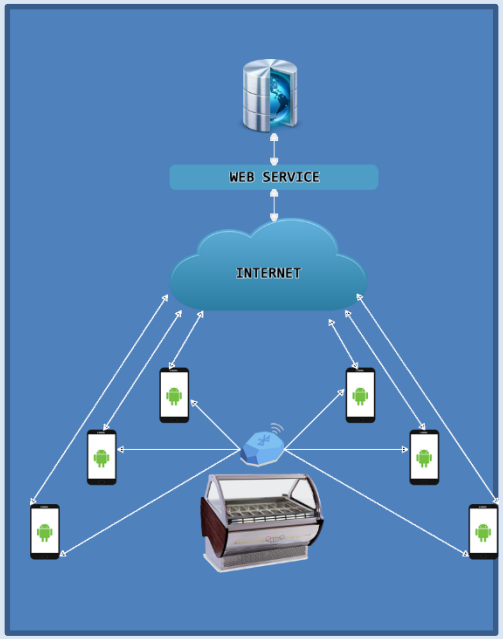
Used Technologies

✓Android Platform

✓BoniBeaconSDK.jar

✓NodeJS

✓MongoDB









Uygulamayı hazırlayan Ergun ALATEŞ

Lucky Beacon

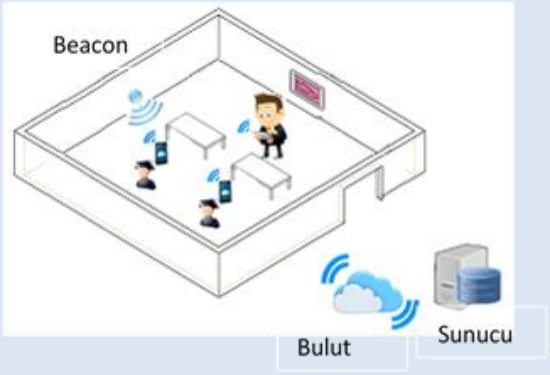
- The application code

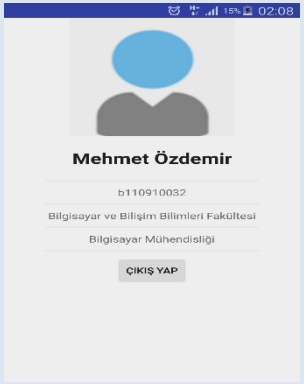
```
BoniBeaconManager.getInstance(context).registerRangingListener(new BoniBeaconListener() {
    @Override
    public void onNearestRangedBeacon(BoniRegion boniRegion, BoniBeacon boniBeacon) {
        if (boniBeacon.getMinor().equals("0002") && boniBeacon.getRssi() > -70) {
            //JSON object oluşturulyor
            try {
                JSONObject jsonObject = new JSONObject();
                jsonObject.accumulate("phone", file.getString("PlaceControl", null));
                jsonObject.accumulate("address", boniBeacon.getDeviceAddress().toString());
                String json = jsonObject.toString();
                //Sunucu ile bağlantı kurulup servis de kullanıcı kontrolü yapılıyor
                String result = new ServiceGetCaller(json).execute(Connection.getWebServiceURL() +
                                                                   "campaignControl").get();
                if (new JSONObject(result).getString("result").equals("LUCKY")) {
                    notifications = new Notifications(getApplicationContext(),
                                    "Şanslı Kişi Sizsiniz", "Tüm ürünlerimizde %50 İNDİRİM");
                    notifications.sendNotification();
                } else {
                    if (region != null) {
                        BoniBeaconManager.getInstance(context).stopRangingBeacons(region);
                    }
                }
            } catch (InterruptedException e) {
                e.printStackTrace();
            } catch (ExecutionException e) {
                e.printStackTrace();
            } catch (JSONException e) {
                e.printStackTrace();
            }
        }
    }
```

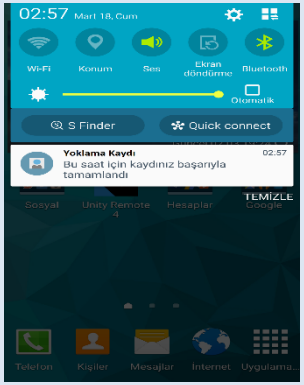
Uygulamayı hazırlayan Ergun ALATEŞ

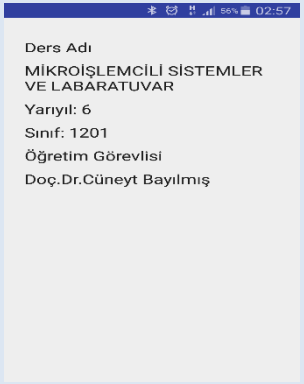
Student Attendance

- The follow-up procedures can be determined according to the length of the course attendance by the automatic student polling system via beacons and smartphones placed in the classes.











Devam Takip Sistemi

Ders Bilgileri

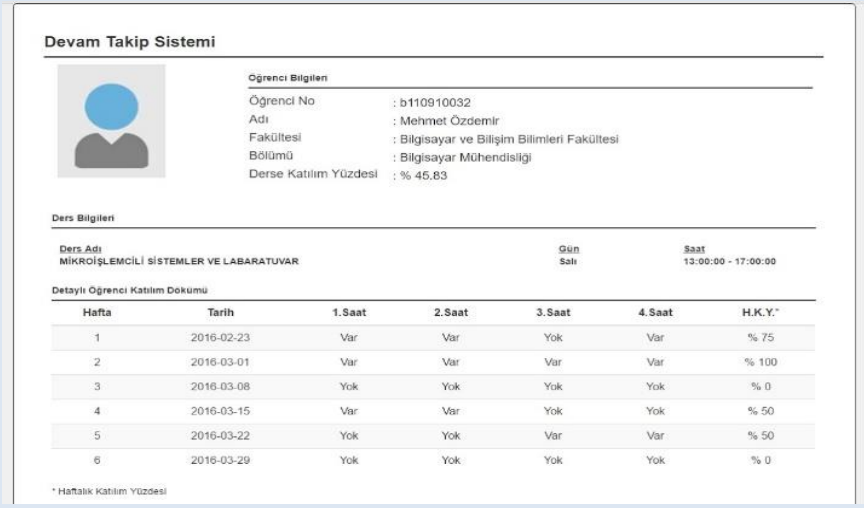
Adı	: MİKROİŞLEMCİLİ SİSTEMLER VE LABARATUVAR
Öğretim Görevlisi	: Doç.Dr. Cüneyt Bayılmış
Yarıyıl	: 6
Gün	: Salı
Saat	: 13:00:00 - 17:00:00
Sınıf	: 1201
Hafta Durumu	: 6 / 14

Ders Açıklaması

Bu derste, günümüz bilgisayar teknolojileri içerisindeki programları işleyen ve çalıştıran mikroişlemcili sistemler hakkında bilgi verilmesi; endüstride yaygın kullanılan 8085 mikroişlemci ve 8051 denetleyici mimari yapılarını ve kodlarını inceleyerek, laboratuar ortamında uygulamalı örnekler ile öğrencilerin teorik ve pratik bilgiler kazanması hedeflenmektedir.

Dersi Alan Öğrenciler

Öğrenci No	Öğrenci Adı	Katılım Durumu
b110910032	Mehmet Özdemir	% 45.83
b110910033	Onur Güneş	% 83.33
b110910024	Zafer Erdoğan	% 0



Devam Takip Sistemi

Öğrenci Bilgileri

Öğrenci No	: b110910032
Adı	: Mehmet Özdemir
Fakültesi	: Bilgisayar ve Bilişim Bilimleri Fakültesi
Bölümü	: Bilgisayar Mühendisliği
Derse Katılım Yüzdesi	: % 45.83

Ders Bilgileri

Ders Adı	Gün	Saat
MİKROİŞLEMCİLİ SİSTEMLER VE LABARATUVAR	Salı	13:00:00 - 17:00:00

Detaylı Öğrenci Katılım Dökümü

Hafta	Tarih	1.Saat	2.Saat	3.Saat	4.Saat	H.K.Y.*
1	2016-02-23	Var	Var	Yok	Var	% 75
2	2016-03-01	Var	Var	Var	Var	% 100
3	2016-03-08	Yok	Yok	Yok	Yok	% 0
4	2016-03-15	Var	Var	Yok	Yok	% 50
5	2016-03-22	Yok	Yok	Var	Var	% 50
6	2016-03-29	Yok	Yok	Yok	Yok	% 0

* Haftalık Katılım Yüzdesi

Kaynak: C. Bayılmış, M. Özdemir, “Bluetooth Düşük Enerji Teknolojisine Sahip İşaretçi ve Akıllı Telefon Temelli Öğrenci Yoklama Sistemi”, Gazi Üniversitesi Bilişim Teknolojileri Dergisi, 9 (3), 249–254, 2016.

References

- C. Bayılmış, M. Özdemir, “Bluetooth Düşük Enerji Teknolojisine Sahip İşaretçi ve Akıllı Telefon Temelli Öğrenci Yoklama Sistemi”, Gazi Üniversitesi Bilişim Teknolojileri Dergisi, 9 (3), 249–254, 2016.

- Beacon Developers
  - Apple: iBeacon for developers, https://developer.apple.com/ibeacon/
  - Estimote beacon, https://estimote.com
  - Blesh beacon, https://www.blesh.com
  - Boni beacon, https://www.boni.me
  - Phoneist beacon, https://www.phoneist.com
  - Google beacon eddystone, https://developers.google.com/beacons/

- BLE Chip Manufacturers
  - Bluegiga BLE113, https://www.bluegiga.com/en-US/products/ble113-bluetooth-smart-module/